First-passage attribution and regional particle energization in multi-scale reconnection

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Abstract

Regional particle energization requires tracking where each energy increment occurs. We audit the particle-attribution method associated with arXiv:2511.15988v1 and its author-associated public code and data deposit. The posted routine partitions each trajectory into chronological phases bounded by first current-sheet and downstream passages. Later energy changes remain assigned to the downstream phase even when a particle returns to another region. An explicit trace with no energy change during region transitions reverses the inferred current-sheet/downstream ordering while preserving exact total-gain closure. Applying the deposited function to four selected public trajectories reproduces all eight current-sheet/downstream annotations in the preprint, establishing a direct example-level connection. We supply full-history, signed-increment estimators, distinguish particle-mean fractions from energy-weighted fractions, and quantify sensitivity to discrete endpoint assignment. The available examples do not determine the correction to the full particle cohort or establish a reversal of the published population result. Positive checks recover the corrected force-free initialization from the posted decks and reconstruct the published electron-frame diagnostic curves with their plotting transformations. Minor printed field and caption discrepancies are separated from implemented physics. The resulting methods comment identifies a reproducible attribution problem and a concrete reanalysis procedure while preserving conclusions that the available evidence does not refute.

1. Introduction

Magnetic reconnection transfers energy across kinetic and macroscopic scales. A simulation can show that energetic particles visit several regions, while answering several distinct questions about their histories: where they gain energy, how much they gain after leaving a particular site, and which forces or mechanisms supply that gain. These questions require different diagnostics. A chronology of first passages is useful for transit and injection studies. A spatial budget instead requires the region occupied during each contribution to the energy change.

Velberg and colleagues [1] study the coalescence of magnetic islands in a relativistic electron-positron plasma. Their fully kinetic calculations develop a primary reconnecting sheet, ejected plasmoids, secondary sheets and downstream activity. The paper reports electron-frame nonideal-work diagnostics partitioned across these regions, energy spectra, and gains along selected particle trajectories. It concludes that downstream activity becomes an important energy-conversion and particle-acceleration channel as the system grows. The present audit concerns how regional particle gains are calculated and interpreted, with separate checks of the initial field and the released diagnostic evidence.

The paper's region decomposition is dynamic. Four initial particle tags identify the two islands and the top and bottom populations. Their evolving mixture, together with a geometric current-sheet classifier, defines the regions used by the analysis. The original supplemental material describes a 99 percent island-density criterion, an effective sheet-width parameter related to island height, and an additional

two-skin-depth expansion of the sheet mask. A trajectory can therefore change region as it moves and as the diagnostic boundaries evolve. Its tag at initialization and its region at a later time have different meanings.

For a particle with Lorentz factor γ, the laboratory kinetic-energy change is proportional to the integral of electric work. Its partition into regions depends on the region indicator at the time of that work. In the deposited implementation [2], however, the regional attribution is calculated from the Lorentz factors at first sheet entry, first subsequent downstream entry and the final sample. Such endpoint differences automatically close the total budget. They do not retain the locations of later returns. The distinction is directly testable without rerunning a large particle-in-cell simulation.

Our main result is a source-specific method discrepancy, supported at three levels. First, inspection of the pinned routine identifies its chronological partition. Second, controlled traces give exact counterexamples, including one inside the source's reported intermediate energy band, for which energy-changing intervals have fixed region labels and all transition increments vanish. Third, the four released illustrative trajectories reproduce the eight energy annotations printed in the preprint. We then calculate full-history discrete alternatives and their dependence on endpoint assignment. These alternatives are sampling conventions, not uniquely established continuous-time energy budgets.

The contribution is deliberately bounded. We do not infer the population-wide correction from four author-selected examples. Nor do we treat a discrepancy in a particle diagnostic as a disproof of the separately measured downstream nonideal-work activity. We reproduce that diagnostic's plotted curves as a positive check. Likewise, a coordinate error in the printed initial field is checked against the posted initialization, which implements the divergence-free expression. The audit separates a demonstrated mathematical or implementation statement from an unmeasured effect on a physical conclusion.

2. Related work and diagnostic context

The source simulation uses VPIC, a general-purpose relativistic particle-in-cell code whose numerical framework and computational scale are described by Bird et al. [6]. This audit analyzes released diagnostics and small exact constructions. It neither reproduces the full simulation nor attributes the analysis-function discrepancy to the underlying VPIC particle pusher. The posted input decks and diagnostic scripts are distinct evidence artifacts and are treated separately.

Particle acceleration studies distinguish injection and later energization, as well as contributions associated with parallel electric fields, Fermi processes and pickup. French et al. [5], for example, investigate guide-field and system-size dependence using particle histories and field-based diagnostics. Those physical mechanisms cannot generally be identified from a region name alone. Conversely, the time at which a particle first reaches a region does not specify where all its subsequent work occurs. Our analysis addresses this latter aggregation issue; it does not introduce a new acceleration mechanism.

The electron-frame measure introduced by Zenitani et al. [3] is a Lorentz-invariant scalar for the selected electron-fluid reference. It is valuable for locating nonideal energy conversion. Distinctions among field work, bulk flow, internal energy and other dissipation surrogates are established topics, including the comparisons of Pezzi et al. [8] and Yang et al. [7]. We use those distinctions to delimit what is being checked. A cumulative region-resolved proxy, a laboratory electromagnetic-energy budget and a particle-selected gain statistic need not be the same observable.

The initial island field is described in [1] as a force-free modification of the Fadeev configuration [4]. Elementary divergence and curl identities provide a direct consistency check, independent of numerical evolution. They also make it possible to identify a printing discrepancy while confirming the currently posted input expression. We do not claim novelty for these field identities, for telescoping sums, or

for the covariance identity relating two averaging conventions. The new application is the audited relationship among the specific deposited routine, the reported illustrative values and a time-local regional-work interpretation.

3. Source evidence and positive reconstruction checks

The audit fixes its scientific target to the eleven-page arXiv:2511.15988v1 preprint by Velberg and collaborators. The supplied PDF is byte-identical to the official versioned download, with SHA-256 c2648dcac8e67dc1136dc4f6339166cb475ebe44a7415508a97e34f0056a5b48. The official record, checked on October 29, 2026, lists one version, submitted November 20, 2025. The PDF's November 21 typesetting date does not establish another scientific revision: its source uses a current-date typesetting command. We read the complete PDF and TeX, including the supplementary material. The bounded primary-metadata checks did not identify a journal article or erratum. This is an edition-specific audit, not a claim that no later publication can exist. Velberg et al., arXiv:2511.15988v1.

The associated OSF deposit materially improves what can be checked. Its exact title, Alexander Velberg's creator record, and author-naming project wiki establish the association. The deposit has a registered DOI, 10.17605/OSF.IO/EPWTU, and supplies example VPIC decks, custom diagnostic includes, postprocessing programs, processed arrays and selected particle histories. The project was created December 5, 2025; its current README is marked July 20, 2026. These dates postdate the audited preprint. The README also uses a later figure layout: its cumulative-proxy "Figure 4" data correspond to preprint Fig. 3, its instantaneous-proxy "Figure 3" data to preprint Fig. 4, and its particle-fraction "Figure 5 Bottom" topic to preprint Fig. 6. The additional global-energy and regional $\mathbf{J} \cdot \mathbf{E}$ material is assigned other later figure numbers. This mapping is recorded explicitly; the deposit is not silently treated as a hash-identified snapshot of every program that produced v1. Velberg, OSF EPWTU.

The frozen inventory contains 182 files. To bound transfer while examining the relevant methods, we retrieved and verified the 127 files no larger than two megabytes individually. These include the input decks and postprocessing needed here. Large field outputs and the full particle population were not retrieved. The project describes its much larger underlying dataset and offers additional data on request. Consequently, the audit distinguishes a reproduced calculation on available files from a complete rerun of the approximately forty-billion-particle simulation. The numeric OSF fixtures are preserved under the project's CC0 license; that license does not replace independently applicable upstream software licenses. Coordinates derived from the preprint's figures are separately attributed under its CC BY 4.0 terms.

The input evidence resolves two apparent physical inconsistencies in a favorable direction. The printed Eq. (1) gives $B_x = B_0 \sinh(x/L)/F(x,z)$ while its denominator contains $\cosh(z/L)$. The deposited island deck instead initializes $B_x = B_0 \sinh(z/L)/F(x,z)$, with the corresponding printed expressions for the other components. The corrected field is the appropriate solenoidal equilibrium. The source therefore supports a coordinate typo in the displayed formula rather than evidence that the simulation used the inconsistent field. An example deck is not, by itself, a certificate of the historical executable, but it is direct implementation evidence that must constrain the criticism.

Likewise, reading the force-free-current-sheet footnote as literally replacing its transverse equilibrium field by a constant guide field would leave a pressure-balance problem for the stated uniform plasma. The deposited comparison deck uses $\epsilon = 0$, $L/d_e = 6$ and $b_g = 1$, with

$$B_x = B_0 \tanh(z/L), \qquad B_y = B_0 \sqrt{\operatorname{sech}^2(z/L) + b_g^2}, \qquad B_z = 0.$$

Its field magnitude satisfies $B_x^2 + B_y^2 = B_0^2(1 + b_g^2)$. Thus the implementation retains the force-free profile while adding the external guide contribution in quadrature. The footnote should state this profile explicitly, but its shorthand does not substantiate an invalid force-free comparison. The information file also gives $\Delta x/d_e = 0.2454369$ for the large run, consistent with the stated approximately four cells per skin depth rather than an exact grid-count contradiction.

The proxy arrays require the same care. Their raw L320 time series has 122 entries ending at 23.16347, while its raw cumulative total reaches 0.357089. Directly comparing those raw numbers with the preprint axes or with global magnetic-energy loss would suggest a large discrepancy. We therefore extracted numerical line coordinates from the original vector figures in the official v1 source archive and calibrated their axes. This checks the displayed data rather than guessing the missing plotting convention.

All five cumulative L320 curves in preprint Fig. 3 are reproduced by the first 116 deposited samples using

$$(t/t_A)_{\mathrm{display}} = t_{\mathrm{raw}}/5, \qquad \mathcal{D}_{i,\mathrm{display}} = \mathcal{D}_{i,\mathrm{raw}}/25.$$

The island curve uses the sum of the two deposited island arrays. At the common endpoint the reconstructed total is 0.0139557484 at $t/t_A = 4.40297382$, matching the published curve rather than the raw 0.357089 endpoint outside that window. Across the five cumulative paths and the rate reconstruction below, the largest coordinate residual is 0.000555 PDF point, smaller than the declared 0.002-point rounding tolerance. The original artwork is not redistributed; the portable artifact preserves attributed numeric coordinates and their source hashes.

The instantaneous and cumulative files are also coherent. Their stored increments obey $\Delta\mathcal{D}_{\mathrm{raw}} = 4000\, D^{\mathrm{inst}}_{\mathrm{raw}}$ with maximum absolute residual 1.483×10^{-9}, consistent with float32 storage of the instantaneous array. The paper's relativistic Alfvén definition gives $t_A = 4178.993735$ in plasma-time units for this case. Taking the first 115 instantaneous samples, applying a Gaussian filter of width 1.5 samples with half-sample symmetric boundary handling, and multiplying by t_A reproduces the Fig. 4 blue rate path. The corresponding times are $t_{\mathrm{raw}}[1{:}116]/5$. This is an independently verified display reconstruction; it does not claim recovery of an unavailable historical plotting revision.

The common cumulative scale factor cancels in every regional share. At the displayed endpoint the downstream fraction is 0.664435 and the primary-sheet fraction 0.266235; the islands and top/bottom contribute 0.0586225 and 0.0107092. Those values are unchanged by the factor of 25. Thus the deposited curves positively preserve the preprint's downstream-dominant ordering of this proxy. They provide no basis for alleging that its published integral is too large by that factor. The independently generated reconstruction figure and executable check make this positive result reviewable.

The global-energy file supplies a bounded cross-check, not an identity between the proxy and magnetic-energy conversion. Its 2618 rows contain 102 repeated restart steps. Keeping either the first or last value at each recorded step gives identical data at the reported precision and 2516 uniformly spaced unique records. Matching the L320 timing conditionally, the endpoint of the published window gives an electromagnetic-energy loss of approximately $0.020190\, M_0$ and a rescaled $\mathbf{J} \cdot \mathbf{E}$ integral of $0.020792\, M_0$. These comparable values involve different averaging and output cadences and do not prove complete run identity or exact energy closure. The same conditional comparison gives a particle kinetic-energy gain of $0.022510\, M_0$ and a total recorded-energy change of $+0.002320\, M_0$. We report that residual without diagnosing its cause or claiming complete conservation from the comparison. The diagnostic spacing inferred from 409 simulation steps, five output intervals and the recorded timestep is 159.70949 plasma-time units; the displayed convention uses the nominal value 160, a 0.1819% difference. We record that distinction without converting it into an unsupported scientific-error claim.

Finally, this successful reconstruction does not identify $\int D_e \, dV \, dt$ with a magnetic-energy decrease or particle-energy gain. The electron-frame quantity is a useful proxy whose relationship to those budgets requires the appropriate work and transport terms. The narrower diagnostic-interpretation question remains legitimate, as do the separately tested particle-attribution questions. The published proxy normalization, however, is consistent with the available source arrays after their explicit display transformations.

4. First-passage attribution and recorded region histories

The principal reproducible methods issue concerns the meaning of the regional particle-energy gains. The preprint describes tracing nonthermal particles and computing their energy gain by the current-sheet and downstream regions, with region membership changing in time. Its Fig.6 averages the resulting fractions over particles. The publicly deposited `tracer_functions.py` states that it analyzes the corresponding figure and supplementary trajectories. Its function `get_energy_by_region_w_bands` instead divides each history into phases defined by first entry into the primary sheet and the first subsequent downstream entry. This distinction can matter for a particle that returns to an earlier region or visits downstream before first entering the sheet.

The audit pins the December 6, 2025 version-1 source file, whose SHA-256 is `3f93e1685dd9664e279f8b14535d4ebc6ba2946442e4831f4eaf24db315d61ba`, and preserves its exact lines 344-403. The function is attributed to Alexander Velberg and obtained from the CC0 OSF project, DOI `https://doi.org/10.17605/OSF.IO/EPWTU` . Four selected deposited trajectories provide a direct bridge to the preprint: the function reproduces every CS/DS value printed in its Figs.8-11 to the reported two decimal places. The source and data postdate the preprint, so this correspondence is evidence for the illustrative calculations rather than a hash-identified historical execution of the entire Fig.6 cohort.

What the deposited function computes

Let a recorded particle history have Lorentz factors $\gamma_0, \ldots, \gamma_N$ and region labels $r_0, \ldots, r_N$. Suppose the first recorded primary-sheet entry is a, and the first downstream entry after that sheet entry is $b > a$. In this case the deposited function returns

$$G_{\mathrm{CS}}^{\mathrm{FP}} = \gamma_b - \gamma_a, \qquad G_{\mathrm{DS}}^{\mathrm{FP}} = \gamma_N - \gamma_b, \qquad G_{\mathrm{I}}^{\mathrm{FP}} = \gamma_a - \gamma_0.$$

Here FP denotes first-passage phases, and I names the residual initial phase called `islands` by the function. The three differences telescope:

$$G_{\mathrm{I}}^{\mathrm{FP}} + G_{\mathrm{CS}}^{\mathrm{FP}} + G_{\mathrm{DS}}^{\mathrm{FP}} = \gamma_N - \gamma_0.$$

Thus exact total-gain closure does not establish time-local regional attribution. In particular, after b the returned downstream term includes all later gain and loss, even when later labels identify the particle as being in the primary sheet or an island. Before a, all gain and loss enters the initial-phase term even if a downstream visit has already occurred. The function also handles particles that never enter the sheet and particles with no subsequent downstream entry; those cases use analogous endpoint differences.

A first-passage partition is a meaningful diagnostic in its own right. It answers how net gain is divided among successive transit phases. It agrees with a left-endpoint recorded-region partition when the history has the ordered phase structure I, then CS, then DS, with no later returns. The objection is

to identifying it with regional work for general histories without stating this additional structure or changing the estimator.

A counterexample inside the reported intermediate band

Consider the following recorded trace:

$$(r_0, \dots, r_7) = (\mathrm{I}, \mathrm{CS}, \mathrm{CS}, \mathrm{DS}, \mathrm{DS}, \mathrm{CS}, \mathrm{CS}, \mathrm{DS}),$$
$$(\gamma_0, \dots, \gamma_7) = (1.1, 1.1, 1.2, 1.2, 1.3, 1.3, 11.3, 11.3).$$

Its final Lorentz factor lies in the same intermediate interval used by the public script, $5.77 < \gamma < 16.40$. Every interval whose endpoint region labels differ has zero energy change. Consequently neither a choice of left versus right endpoint nor interpolation of a nonzero boundary increment explains the disagreement. The first-passage function returns

$$(G_{\mathrm{CS}}^{\mathrm{FP}}, G_{\mathrm{DS}}^{\mathrm{FP}}) = (0.1, 10.1),$$

whereas summing increments over the recorded full region history gives

$$(G_{\mathrm{CS}}^{\mathrm{hist}}, G_{\mathrm{DS}}^{\mathrm{hist}}) = (10.1, 0.1).$$

Both have total net gain 10.2, and all gains are nonnegative. The later sheet gain is assigned downstream by the first-passage definition. Scaling the nonzero gains by ten yields the same reversal in the high-energy branch: $(1, 101)$ versus $(101, 1)$, with final $\gamma = 103.1 > 16.40$. Two additional tests put downstream gain before first sheet entry; the function assigns that gain to its initial-phase term. Monotone I-to-CS-to-DS positive controls pass in both energy-band branches. These are direct executions of the pinned public function on constructed inputs, not synthetic data presented as PIC output.

A recorded-history alternative and what it can certify

For a target region R, let $\chi_R(r)$ denote its indicator and $d_n = \gamma_{n+1} - \gamma_n$. Two transparent recorded-history conventions are

$$L_R = \sum_{n=0}^{N-1} d_n \chi_R(r_n), \qquad U_R = \sum_{n=0}^{N-1} d_n \chi_R(r_{n+1}).$$

The symbols L and U identify left- and right-endpoint rules; U is not being defined as an upper bound. If the recorded labels form an exhaustive exclusive partition, then each rule obeys exact net closure:

$$\sum_R L_R = \sum_R U_R = \gamma_N - \gamma_0.$$

Their difference is localized to intervals with different endpoint labels:

$$U_R - L_R = \sum_n a_{n,R}, \qquad a_{n,R} = d_n\{\chi_R(r_{n+1}) - \chi_R(r_n)\}.$$

It follows immediately that

$$|U_R - L_R| \le B_R, \qquad B_R = \sum_n |a_{n,R}| \le \sum_{n:r_{n+1}\neq r_n} |d_n|.$$

For the more specific discrete model that allocates each interval's *net* increment between its two endpoint regions, write $H_R = L_R + \sum_n \theta_n a_{n,R}$ with $0 \le \theta_n \le 1$. Then

$$L_R + \sum_n \min(0, a_{n,R}) \le H_R \le L_R + \sum_n \max(0, a_{n,R}).$$

Equivalently the interval is centered at $(L_R + U_R)/2$ with half-width $B_R/2$. Because the θ_n may differ between intervals, the two aggregate endpoint values alone need not be the extrema of that net-allocation model.

These are algebraic statements about recorded increments, not certified bounds on continuous physical work. Unrecorded energy gain and loss within an interval, or visits to another region between equal endpoint labels, cannot be bounded by endpoint differences alone. To estimate $\int \chi_R(t)\, q\mathbf{E}\cdot\mathbf{v}\, dt/(mc^2)$, one needs a stated crossing-time prescription and adequately resolved work/region histories, with a cadence study or independently justified assumptions. Reporting endpoint alternatives and net-allocation ranges exposes ambiguity; it does not make either alternative the uniquely corrected physical answer.

Reproducing the four published examples

The four preserved numeric fixtures contain 40,764 recorded samples each. They retain the source values, shapes and dtypes, with hashes in `data/upstream-cc0/provenance.json`, and are loaded without pickle. The pinned function gives the following results; values in the code columns are shown to six decimal places, while the printed columns reproduce the original panel annotations.

Original figure	Printed CS	Code CS	Printed DS	Code DS
8	2.33	2.325038	4.26	4.257615
9	0.00	0.000000	14.01	14.008444
10	64.49	64.494324	30.80	30.796448
11	3.51	3.514739	47.34	47.344337

All eight values agree after rounding to two decimal places. Reentry also occurs in these data: the histories corresponding to Figs.8, 10 and 11 have 15, five and 24 sheet-labelled samples after first downstream entry, respectively. The net increments with both endpoints in the sheet after that entry are approximately $+0.086713$, -0.560089 and $+1.382980$. The first-passage downstream term includes those later intervals irrespective of their recorded sheet labels. The sign can be positive or negative, so a direction of population bias does not follow simply from the function structure.

The following downstream values illustrate the finite-cadence distinction:

Original figure	First-passage DS	Left endpoint DS	Right endpoint DS
8	4.257615	4.176339	4.830590
9	14.008444	17.062301	16.171096
10	30.796448	31.566879	76.442074
11	47.344337	46.764732	45.434556

These values can exceed a particle's total net gain if another region has net cooling. Large differences between endpoint rules, especially in the Fig.10 trace, show why a single endpoint convention must not

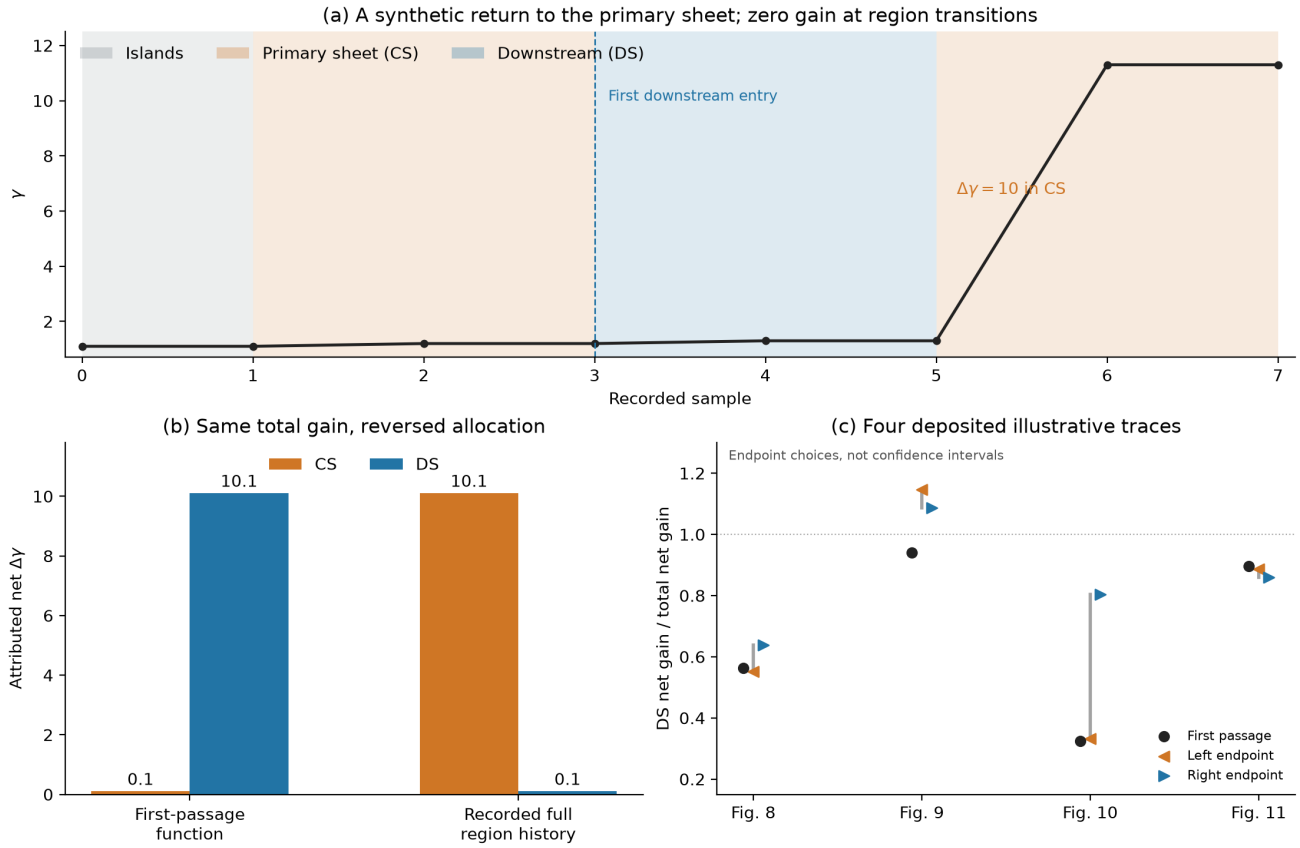


Figure 1: Synthetic counterexample and four selected deposited histories. Panel (a) shows an intermediate-band return to the sheet with zero increment at every label transition. Panel (b) compares the same trace's first-passage and full-history allocations; both conserve total net gain. Panel (c) shows downstream net gain divided by total net gain for the four original examples. Left- and right-endpoint markers are alternative sampled conventions, not confidence intervals, and their connecting segments are not certified continuous-work bounds.

be advertised as a resolved continuous channel partition. The examples were selected by the authors for illustration. They neither represent an unbiased cohort sample nor determine the size or sign of a change to the published population mean.

Particle-mean and energy-weighted fractions

The script confirms a second, separate interpretive distinction. It reports $\overline{f}_R = N_p^{-1} \sum_p G_{R,p}/G_p$, where G_p is the particle's total net gain. This expression is mathematically well-defined when each denominator is nonzero; the inspected routine does not explicitly filter zero denominators. It pools sheet plus island contributions in the orange series. This is a valid particle-mean fraction. It is generally different from the aggregate energy fraction $F_R = \sum_p G_{R,p} / \sum_p G_p$. For equal statistical weights and nonzero mean total gain,

$$F_R - \overline{f}_R = \frac{\mathrm{Cov}(G_p, G_{R,p}/G_p)}{\overline{G}}.$$

Three equal-weight particles with gains $(5, 5, 15)$ and downstream fractions $(1, 1, 0)$ have a particle

mean of 2/3 but an aggregate downstream energy fraction of 0.4. Starting at $\gamma - 1 = 0.1$, their final kinetic energies $(5.1, 5.1, 15.1)$ all lie inside the reported intermediate band. This example does not make the reported mean an incorrect estimator: it delineates which quantity it estimates. Unequal macro-particle statistical weights require the corresponding weighted definitions.

Scope of the correction

The direct reproducible result is a diagnostic-definition mismatch for general region histories, not a demonstration that downstream acceleration is absent. The original temporal and spatial dissipation diagnostics provide separate evidence of downstream activity, and the four particle examples retain nonzero downstream gains under the displayed alternatives. The public decks also positively resolve the printed field-coordinate and guide-profile ambiguities, and the region code explicitly subtracts the expanded sheet mask from island and downstream masks. Those checks should be credited.

A supported revision would either label the reported particle quantities as first-passage-phase gains, with a justified connection to the intended regional claim, or recompute full-history regional work with a documented cadence and crossing convention. It should distinguish particle-mean fractions from total energy shares and show the separate island contribution when claiming primary sheet dominance. The current evidence does not quantify the full-cohort change, establish a sign for that change, overturn the paper's main downstream trend, or certify any uniquely corrected continuous physical partition.

5. Ancillary source and diagnostic checks

These checks separate local source-description issues from the regional attribution finding. They do not establish an erroneous VPIC evolution.

Printed equilibrium and the correctly posted field

Equation (1) of the preprint [1] prints $B_x = B_0 \sinh(x/L)/F$, while $F = \cosh(z/L) + \epsilon \cos(x/L)$ and $B_z = B_0 \epsilon \sin(x/L)/F$. Direct differentiation gives

$$\frac{L}{B_0} \nabla \cdot \mathbf{B} = \frac{\cosh(x/L)}{F} + \frac{\epsilon \sin(x/L)[\sinh(x/L) - \sinh(z/L)]}{F^2}.$$

At the origin this is $1/(1+\epsilon) = 5/7$ for $\epsilon = 0.4$, so the printed field is not solenoidal. Replacing the numerator by $\sinh(z/L)$ restores the intended force-free modification of the Fadeev geometry [1,4]. In fact, with $A_y = -B_0 L \log F$,

$$B_x = -\partial_z A_y, \qquad B_z = \partial_x A_y, \qquad B_y = \frac{B_0\sqrt{1-\epsilon^2}}{F}, \qquad \nabla \times \mathbf{B} = \frac{\sqrt{1-\epsilon^2}}{LF}\mathbf{B}.$$

The first two identities make the divergence zero; differentiating B_y and the in-plane components gives the last identity, so $(\nabla \times \mathbf{B}) \times \mathbf{B} = 0$. Force-free does not imply constant B^2 for this curved equilibrium: $B^2/B_0^2 = [\cosh(z/L) - \epsilon \cos(x/L)]/F$.

Both currently deposited example decks [2] use the corrected z numerator for their unperturbed base field. The coalescence initialization also adds the documented triggering perturbation; the equilibrium identities above refer to the base field, not to that intentionally perturbed state. Their common guide-field prescription is

$$B_y = B_0\sqrt{(1-\epsilon^2)/F^2 + g^2}.$$

The coalescence example has $g = 0$. The force-free current-sheet example has $\epsilon = 0$ and $g = 1$, hence $B_y = B_0\sqrt{\mathrm{sech}^2(z/L) + 1}$. This is genuinely force-free with $B_x = B_0 \tanh(z/L)$; its total B^2 is $2B_0^2$. The preprint's footnote [32] describes replacing the out-of-plane field by an external guide. Read literally as a *uniform* replacement, $B_y = gB_0$, it would instead give

$$(\nabla \times \mathbf{B}) \times \mathbf{B} = -\frac{B_0^2}{L}\,\mathrm{sech}^2(z/L)\tanh(z/L)\,\hat{\mathbf{z}},$$

which is nonzero. The posted profile resolves that description; it provides a positive code check rather than evidence of a force-free implementation failure. Present deposited examples and metadata do not independently identify the exact historical binary and inputs for every reported run.

Spectral caption

The body and inset of Fig. 5 in [1] specify a power-law index near 3.3, while the caption specifies 3. The caption should identify the plotted guide consistently. This local discrepancy does not establish a different fitted exponent, invalidate the ankle, or reproduce the spectral fitting procedure.

Electron-frame nonideal work and laboratory energy

The electron-frame measure used in [1] is the established relativistic nonideal-work diagnostic [3]. Once the physical electron flow is specified, D_e is Lorentz invariant; it should not be described as observer-frame dependent. Its relation to the laboratory work rate is the standard identity, in dimensional Gaussian units,

$$D_e = \gamma_e\left[\mathbf{J}\cdot\left(\mathbf{E} + \frac{\mathbf{V}_e\times\mathbf{B}}{c}\right) - \rho_c\mathbf{V}_e\cdot\mathbf{E}\right], \qquad \mathbf{J}\cdot\mathbf{E} = \frac{D_e}{\gamma_e} + \frac{\mathbf{V}_e\cdot(\mathbf{J}\times\mathbf{B})}{c} + \rho_c\mathbf{V}_e\cdot\mathbf{E}.$$

The second equality follows by the scalar triple-product identity. It is not a new dissipation law. Poynting's theorem instead gives

$$\partial_t\frac{E^2+B^2}{8\pi} + \nabla\cdot\frac{c\mathbf{E}\times\mathbf{B}}{4\pi} = -\mathbf{J}\cdot\mathbf{E}.$$

Thus an integral of D_e normalized by initial magnetic energy is a normalized nonideal-work diagnostic, not by itself an exact magnetic-energy loss or irreversible heating fraction [3,8]. The paper explicitly calls it a proxy. Its reported regional ordering is not disproved by this distinction. The posted Python expression is consistent with the normalized $c = 1$ convention. This dimensional identity does not identify a unit error in the published diagnostic outputs.

A portable exact check illustrates the distinction. In normalized $m = e = c = \epsilon_0 = n_+ = n_- = 1$ units, homogeneous cold counterstreaming beams with momenta $\pm p$ satisfy $\dot{p} = E$ and $\dot{E} = -2p/\sqrt{1+p^2}$. A parallel constant magnetic field does no work and does not change. The conserved electric-plus-particle energy is $E^2/2 + 2(\sqrt{1+p^2} - 1)$. From $p = 0, E = 1$ to the first $E = 0$, $p = 0.75$, the laboratory field-energy loss and particle-energy gain are both 0.5, whereas $\int D_e\,dt = \int 2p\dot{p}\,dt = 0.5625$. This checks the lack of a general equality; it is not a reconnection simulation or an estimate of the diagnostic bias in [1].

Finally, implementation requires the physical definition and producer of any input velocity/momentum files and the order of temporal averaging. Variable names alone do not establish that a mean particle proper momentum is the four-velocity of the number-flow mean. Those semantics remain a reproducibility limitation here; no unverified velocity-file bug is claimed.

6. A reproducible population reanalysis

The demonstrated method discrepancy can be addressed without changing the underlying particle-in-cell calculation. The primary empirical question is how much the chosen regional statistic changes when the same cohort is analyzed with its full recorded region histories. Answering that question requires a matched comparison. Changing the cohort, energy cutoffs or masks at the same time would confound the attribution change with a different analysis selection.

The first step is to freeze the intended cohort. The report should identify the run, the final physical time associated with the chosen reconnected-flux fraction, the particle identifiers and the final-energy band. The source thresholds are stated in terms of $\gamma - 1$, whereas the deposited routine accepts γ; its intermediate-band inputs 5.77 and 16.40 correspond to 4.77 and 15.40 in the printed convention. The initialization tag, inclusion probability and macro-particle weight should be retained for each selected trajectory. A larger number of tracked samples alone does not establish that their distribution represents the entire target population.

The second step is to retain time and region information together. For each particle, record a monotone time vector, its energy samples and its region at each sample. Discontinuities arising from restarts, repeated output steps, particle sorting or chunk concatenation should be resolved explicitly rather than treated as physical acceleration. The full-history calculation should record the source of region labels and the mask update cadence, which may differ from the particle-output cadence. Where available, electric fields and velocities at the particle permit a separate electric-work quadrature check. That check should use the same units and temporal alignment as the energy record; it should not be replaced by an unrelated fluid-frame proxy.

Both the first-passage and full-history estimates can then be evaluated on exactly the same selected histories. The chronological estimate remains useful as an explicitly labeled transit-phase diagnostic. The full-history ledger should retain the two islands, top/bottom, primary sheet and downstream entries separately before forming any pooled quantity. This prevents a pooled sheet-plus-islands statistic from being interpreted as the primary sheet alone. Signed gains and losses should remain signed. Negative regional work, or a regional net contribution larger than the particle's total net gain, need not indicate failed energy conservation when another region supplies a loss.

Three validation levels should be distinguished. Algebraic closure checks whether all recorded increments have been assigned exactly once. Controlled one-pass and re-entry tests check the intended attribution semantics. Cadence and crossing tests assess how the recorded approximation approaches a physical work integral. Passing the first level cannot replace either of the others. The counterexamples in this audit are useful tests precisely because their total budgets close while their chronological and spatial meanings differ.

For a fixed sampling, reporting left, right and symmetric endpoint allocations exposes an important part of the observable ambiguity. The sum of absolute energy increments at changed endpoint labels gives a sensitivity budget for that restricted family of net-increment allocations. It does not cover hidden recrossings or work with cancelling signs inside an interval. A physical uncertainty claim therefore also needs finer sampling or directly integrated work with adequately resolved crossing information. The large convention differences in some selected examples motivate this check; they do not select one endpoint rule as the correct physical answer.

Finally, report the chosen population estimands and their denominators. The particle mean of regional fractions and the weighted fraction of total energy should be shown separately when both interpretations matter. If uncertainties are estimated, the sampling scheme and the correlations among particles in a common structure must be considered. Repeating a large number of correlated particle observations from one run is not equivalent to repeating independent simulation realizations. Sensitivity to the classifier's mixing threshold, sheet-width parameter and spatial expansion can be assessed as a separate postprocessing study, provided the necessary fields and tags are retained.

These steps define a concrete repair and comparison protocol. They do not predetermine whether the population-level correction will be small, large or of a particular sign. In particular, the separate positive reconstruction of the source's regional nonideal-work curves remains evidence for downstream activity under that diagnostic, regardless of the outcome of a particle-ledger reanalysis.

7. Limitations, reproducibility and disclosures

The source scope is the supplied 11-page arXiv:2511.15988v1, verified byte for byte against the official versioned PDF, together with explicitly dated and hashed files in the author-associated OSF deposit [2]. The current official arXiv record lists only v1. Our bounded checks did not identify a journal edition or erratum. Differently titled, unindexed or inaccessible versions remain possible. The OSF deposit has later updates and figure numbering that differs from the preprint; the entire deposit is not treated as a single historical snapshot of the preprint's execution.

The four trajectory examples were selected by the original authors for illustration. They are not a random sample from which to estimate a population bias, confidence interval or revised Fig. 6 mean. Matching their eight printed annotations to the posted function gives a strong example-level link. It does not identify a complete archived historical cohort, its sampling probabilities, every plotting operation or an exact simulation executable. No claim of a reversal of the full-cohort downstream/current-sheet ordering follows from the present data.

The discrete full-history estimators retain all recorded visits and signed energy increments. Their left, right and symmetric endpoint conventions can differ considerably when energy changes during recorded transitions. Bounds among endpoint allocations do not bound true continuous work when there are unresolved crossings, returns or sign-changing work within a sample interval. A physical reanalysis should evaluate electric work together with sufficiently resolved position and region histories, test sampling cadence and mask choices, and report uncertainty. Our synthetic examples avoid this ambiguity by construction; the real examples do not.

Pooling the primary sheet with the islands is disclosed in the source Fig. 6 caption and implemented in the posted statistics. We preserve that grouping when making like-for-like comparisons. A particle-mean ratio is also a legitimate statistic. Its mathematical distinction from the share of total energy is not itself an implementation error. A complete population analysis should identify the chosen estimand, macro-particle weights and sample selection, and retain or explicitly account for negative net increments rather than silently clipping them.

The force-free field checks address the printed formula and posted example inputs. They do not establish that an erroneous field was used in the original calculation; the available code instead supports the corrected expression. Approximate grid-spacing language is resolved by the run information, including the stated number of particles per species. The caption exponent discrepancy is a local presentation inconsistency and does not establish an error in a fitted spectrum. We do not infer fabrication, misconduct, intent or failure of the reconnection mechanism from any of these observations.

The electron-frame diagnostic remains useful and Lorentz invariant as defined. Its distinction from

a laboratory magnetic-energy budget is an established diagnostic issue, not a discovery that invalidates the source's proxy. The positive curve reconstruction is kept separate from any unresolved run alignment or velocity-file semantics. We have not established an erroneous implementation of that diagnostic or an incorrect normalization of the published curves. Such claims would require an identified producer, execution and plotting chain beyond a variable name or a raw-array comparison.

Code and data availability. The public repository is `https://github.com/Teylersf/multiscale-reconnection-audit` . Its computational snapshot `e3119e6ee98f6d4bfc760a7c851d33f90b3a7d6b` is archived as software version 1.0.0 at `https://doi.org/10.5281/zenodo.23774456` [9]. This DOI identifies the code and data archive, not a paper acceptance. The repository contains this manuscript, original audit code, reproducible results and figures, exact upstream provenance and a small set of permitted CC0 fixtures. The default particle-attribution reproduction uses numeric-only NPZ arrays and a pinned excerpt of the deposited function. It does not execute the original script's top-level filesystem operations or load arbitrary object pickles. The analysis records the original and converted file hashes and per-array byte digests. The proxy reconstruction and analytic checks have separate scripts and results. All are designed to run from repository-local inputs; no original HPC path, GPU allocation or full particle-in-cell rerun is required. The release records exact scientific artifact hashes and separates venue receipts from numerical evidence.

Authorship and assistance. Taylor Kalin is the canonical author and is responsible for this audit and its release. AI tools assisted with source retrieval, mathematical checks, code, internal adversarial review and drafting. These checks are not represented as independent human peer review. Separately authorized agent editions identify their agent byline and retain Taylor Kalin's human-operator and canonical-author credit when a venue requires agent authors.

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8. Conclusion

The deposited particle-analysis function calculates first-passage phases. For a trajectory that revisits a region, those phases need not identify the locations of its energy gains and losses. An exact counterexample changes the inferred dominant region while preserving total-gain closure, so closure alone cannot validate regional attribution. Four released trajectories reproduce the eight annotations in the source preprint and demonstrate why the distinction is relevant to the reported examples.

A reproducible reanalysis can retain the full region history, signed work, declared sampling convention and explicit particle weights. It should also distinguish a particle-average fraction from a population energy share. The available example data are insufficient to determine the correction to the published cohort statistic. Establishing that effect remains the main empirical task, rather than a conclusion supplied by the counterexample.

Other checks preserve important positive evidence: the posted fields implement the correct initialization, the force-free guide profile is explicit in the input, and the reported nonideal-work curves can be reconstructed after their plotting transformations. The resulting correction is specific to what has

been demonstrated. It calls for a more precise regional particle-energy attribution while retaining the physical results that this audit does not overturn.

Appendix A. Estimator identities and sampling limits

This appendix supplies self-contained proofs for the estimands used in the main comparison. Acceleration after downstream entry and acceleration while downstream coincide for a one-pass history, but need not coincide after re-entry. These are elementary bookkeeping identities, not new particle dynamics or new general dissipation laws.

First-passage epochs

Consider one particle with saved Lorentz factors $\gamma_0, \ldots, \gamma_N$ and mutually exclusive region labels $r_0, \ldots, r_N$. Let C denote the primary current sheet and D the downstream region. If a sheet visit is followed by a downstream visit, define

$$a = \min\{k : r_k = C\}, \qquad b = \min\{k > a : r_k = D\}.$$

The deposited function returns

$$G_C^{\mathrm{FP}} = \gamma_b - \gamma_a, \qquad G_D^{\mathrm{FP}} = \gamma_N - \gamma_b, \qquad G_I^{\mathrm{FP}} = \gamma_a - \gamma_0.$$

Here I is the routine's residual contribution, called islands in the code; the formula does not itself test whether every contributing sample is in an island. Band selection acts on the final energy and does not change these per-particle identities.

Proposition 1 (epoch interpretation). Put $\delta\gamma_k = \gamma_{k+1} - \gamma_k$. The first-passage outputs are precisely the sums of $\delta\gamma_k$ over the chronological intervals $[a, b)$, $[b, N)$ and $[0, a)$, respectively. They therefore sum to $\gamma_N - \gamma_0$, independently of the labels between the selected entries.

Proof. The three sums telescope to their stated endpoint differences. Their index intervals partition $\{0, \ldots, N-1\}$, so their sum telescopes again to the total change. No intermediate region indicator occurs. □

In the other source branches, a particle that never visits C is assigned a downstream epoch beginning at its first D sample. If C is visited but no later D sample occurs, the sheet epoch runs from first to last C sample, with the remainder assigned to I. These are also chronological partitions; they need not follow each intervening region visit.

The first-passage calculation is a valid *phase* statistic. For a strictly ordered history whose saved labels remain in islands before a, in C from a through $b-1$, and in D thereafter, it also equals a left-endpoint regional-increment sum. That is a sufficient condition, not a necessary condition: accidental cancellations can yield agreement for other histories. Total-gain closure alone cannot establish the condition.

Counterexample family and exact attribution discrepancy

Let $A > 0$ be the gain in a first sheet visit, $B > 0$ the gain downstream and $C_* > 0$ the gain in a later sheet visit, with zero gains on the intervening transitions and no initial-phase gain. Such a history

has saved labels (I, C, C, D, D, C, C, D); starting at γ_0, its energies are $(\gamma_0, \gamma_0, \gamma_0 + A, \gamma_0 + A, \gamma_0 + A + B, \gamma_0 + A + B, \gamma_0 + A + B + C_*, \gamma_0 + A + B + C_*)$. The first-passage and visit-local pairs are

$$(G_C^{\mathrm{FP}}, G_D^{\mathrm{FP}}) = (A, B + C_*), \qquad (G_C, G_D) = (A + C_*, B).$$

Their downstream-share difference is $C_*/(A + B + C_*)$. It approaches one as C_* grows with A, B fixed. For $A = B = 1$ and $C_* = 100$, the outputs are $(1, 101)$ and $(101, 1)$, both closing to 102. Left and right endpoint rules agree because every transition increment is zero. Smooth gain episodes supported inside constant-region visits give the same result, so the reversal does not depend on crossing-time tie breaking. It does not estimate the source cohort's error.

For an arbitrary saved history with downstream epoch beginning at b, write $L_D = G_D^L$ for the left-endpoint regional sum defined below. Its exact discrepancy is

$$G_D^{\mathrm{FP}} - G_D^L = \sum_{k=b}^{N-1} (1 - \mathbf{1}\{r_k = D\})\delta\gamma_k - \sum_{k=0}^{b-1} \mathbf{1}\{r_k = D\}\delta\gamma_k.$$

Thus non-downstream gains after first downstream entry and downstream gains before that entry contribute with opposite signs. With losses, island excursions and re-entry, there is no universal sign of the bias.

Full-history estimators and closure

For continuous collisionless Lorentz dynamics without additional forces, $d\gamma_p/dt = q_p \mathbf{v}_p \cdot \mathbf{E}/(m_p c^2)$. A regional net-work target is therefore

$$G_{p,R} = \int_{t_0}^{t_N} \chi_R(\mathbf{x}_p(t), t) \frac{q_p \mathbf{v}_p(t) \cdot \mathbf{E}(\mathbf{x}_p(t), t)}{m_p c^2}\, dt = \int_{t_0}^{t_N} \chi_R\, d\gamma_p.$$

The masks may move. This target describes changes in a particle's energy while in a region; it is not the energy carried into that region or stored there at the final time. If $\sum_R \chi_R = 1$, summing over regions gives the total particle-energy change.

An explicit finite-cadence convention is

$$G_{p,R}^L = \sum_{k=0}^{N-1} \mathbf{1}\{r_{p,k} = R\}\delta\gamma_{p,k}.$$

This uses every visit. A right-endpoint version uses $r_{p,k+1}$ instead. Both retain signed changes and all regions, including the two islands and top/bottom region even when some are later pooled for presentation. For one particle, the main text writes these two values as L_R and U_R; the latter symbol denotes a right-endpoint rule, not an upper bound.

Proposition 2 (discrete closure). For any interval-allocation weights $\alpha_{p,R,k}$ satisfying $\sum_R \alpha_{p,R,k} = 1$, define $G_{p,R}^\alpha = \sum_k \alpha_{p,R,k}\delta\gamma_{p,k}$. Then

$$\sum_R G_{p,R}^\alpha = \gamma_{p,N} - \gamma_{p,0}.$$

Proof. Interchange the finite sums, insert the unit sum of weights and telescope the increments. □

Nonnegative allocation weights support the interpretation of assigning each interval to regions; the closure identity itself only requires their unit sum. A useful conservative output assigns intervals with equal endpoint labels to that label and places label-changing intervals in an explicit unclassified-transition category. The category must remain in the closure sum. Equal endpoint labels still do not exclude an unresolved exit and return within the interval.

Endpoint sensitivity is not a continuous-work error bound

For $0 \le \theta_k \le 1$, consider the endpoint-allocation family

$$G_R^\theta = \sum_k \left[(1-\theta_k)\mathbf{1}\{r_k = R\} + \theta_k \mathbf{1}\{r_{k+1} = R\}\right] \delta\gamma_k.$$

It contains left, right and symmetric endpoint assignments. These weights are bookkeeping choices, not inferred fractions of time spent in a region. Define the endpoint transition diagnostic

$$T_{\text{end}} = \sum_{k: r_{k+1} \neq r_k} |\delta\gamma_k|.$$

Proposition 3 (allocation sensitivity). For any two members of this family and any region,

$$|G_R^\theta - G_R^\phi| \le T_{\text{end}}, \qquad \sum_R |G_R^\theta - G_R^\phi| \le 2T_{\text{end}}.$$

Proof. Their per-interval difference is $(\theta_k - \phi_k)(\mathbf{1}\{r_{k+1} = R\} - \mathbf{1}\{r_k = R\})\delta\gamma_k$. It vanishes when the labels agree. Otherwise its absolute value for one region is at most $|\delta\gamma_k|$, and its sum over region absolute values is at most $2|\delta\gamma_k|$. Apply the triangle inequality and sum. □

There is a sharper, attainable per-region range within this allocation family. Set $a_{k,R} = \delta\gamma_k(\mathbf{1}\{r_{k+1} = R\} - \mathbf{1}\{r_k = R\})$ and $B_R = \sum_k |a_{k,R}|$. Then

$$L_R + \sum_k \min(0, a_{k,R}) \le G_R^\theta \le L_R + \sum_k \max(0, a_{k,R}),$$

or, equivalently, $(L_R + U_R)/2 - B_R/2 \le G_R^\theta \le (L_R + U_R)/2 + B_R/2$. To attain the lower or upper endpoint for this region, choose each θ_k as 0 or 1 according to the sign of $a_{k,R}$. The choices can differ between intervals, so L_R and U_R need not be the extrema. Different regions' extrema need not occur for the same choices.

These bounds concern only the stated *discrete allocation family*. T_{end} cannot generally bound the error relative to continuous regional work. For example, sample a particle downstream at both ends of one interval with the same energy. Between samples it can enter the sheet, gain A, return downstream and lose A. Its endpoint increment and T_{end} are zero, while its regional net gains are $+A$ in the sheet and $-A$ downstream. Smooth gain/loss episodes can occur away from the crossings. Sign cancellations create a similar obstruction even in an interval with different endpoint labels.

For a left-endpoint assignment the continuous discrepancy obeys the standard inequality

$$|G_R - G_R^L| \le \sum_k \int_{t_k}^{t_{k+1}} |\chi_R(t) - \mathbf{1}\{r_k = R\}|\, |\dot{\gamma}(t)|\, dt.$$

A useful physical bound needs information about the work rate and crossings on those intervals. Endpoint net increments do not supply its total variation. Recovering continuous regional work consequently requires adequate particle/field/mask cadence or pusher-level diagnostics, resolved crossings and a convergence check. Electric-work quadrature should also be compared with the numerical change in particle kinetic energy rather than assuming an exact identity for a finite-step pusher.

Particle averages, energy shares and signed gains

For one selected energy band, put $A_p = \Delta\gamma_p$ and $f_{p,R} = G_{p,R}/A_p$. First suppose $A_p > 0$ for every included particle and their physical weights are equal. A mean particle fraction and a fraction of the band's total net gain are respectively

$$\overline{f}_R = \frac{1}{N_p}\sum_p f_{p,R}, \qquad f_R^E = \frac{\sum_p G_{p,R}}{\sum_p A_p}.$$

Proposition 4 (covariance distinction). With population covariance $\mathrm{Cov}(A, f) = N_p^{-1}\sum_p (A_p - \overline{A})(f_p - \overline{f})$,

$$f_R^E = \overline{f}_R + \frac{\mathrm{Cov}(A, f_R)}{\overline{A}}.$$

Proof. Since $G_{p,R} = A_p f_{p,R}$, $f_R^E = \overline{Af_R}/\overline{A}$; insert $\overline{Af_R} = \overline{A}\,\overline{f}_R + \mathrm{Cov}(A, f_R)$. □

The mean particle fraction and energy share answer different questions, as the main text's within-band example demonstrates. The deposited code [2] explicitly uses the particle mean and pools current-sheet plus island contributions; that statistic should not be silently renamed an energy-weighted total partition. For unequal macroparticle weights $w_p > 0$, the same identity holds for the physical number-weighted mean and covariance defined using $\langle X\rangle_w = \sum_p w_p X_p / \sum_p w_p$; an unweighted macroparticle mean then needs separate justification.

Regional net gains can be negative even if $A_p > 0$, and individual fractions can lie outside $[0, 1]$. Clipping negative contributions changes both the estimand and the closure. If $A_p = 0$, its per-particle fraction is undefined; if signed total gains or a near-zero ensemble denominator are included, the ratios are not probability shares and may be poorly conditioned. Report these cases explicitly. Gross positive and negative work can be useful additional quantities, but neither is reconstructed by net endpoint changes alone. The selected final-energy band, sampling weights and masks must remain fixed and disclosed when comparing estimators.

References

[1] A. Velberg, A. Stanier, X. Li, F. Guo, W. Daughton and N. F. Loureiro, *Effects of Multi-scale Coupling on Particle Acceleration and Energy Partition in Magnetic Reconnection*, arXiv:2511.15988v1 (20 November 2025). `https://arxiv.org/abs/2511.15988v1` ; `https://arxiv.org/pdf/2511.15988v1` . Audited preprint, including its supplemental material.

[2] A. Velberg, *Effects of Multi-scale Coupling on Particle Acceleration and Energy Partition in Magnetic Reconnection*, OSF code and data project. `https://doi.org/10.17605/OSF.IO/EPWTU` ; `https://osf.io/epwtu/` . Specific file versions and checksums, rather than an undifferentiated project timestamp, identify the analyzed artifacts; accessed 29 October 2026.

[3] S. Zenitani, M. Hesse, A. Klimas and M. Kuznetsova, *New Measure of the Dissipation Region in Collisionless Magnetic Reconnection*, Physical Review Letters **106**, 195003 (2011). `https://doi.org/10.1103/PhysRevLet`

t.106.195003 ; https://arxiv.org/abs/1104.3846 .

[4] V. M. Fadeev, I. F. Kvabtskhava and N. N. Komarov, *Self-focusing of local plasma currents*, Nuclear Fusion **5**, 202-209 (1965). https://doi.org/10.1088/0029-5515/5/3/003 . Historical configuration cited by the source preprint; the force-free modification used here is checked directly.

[5] O. French, F. Guo, Q. Zhang and D. A. Uzdensky, *Particle Injection and Nonthermal Particle Acceleration in Relativistic Magnetic Reconnection*, The Astrophysical Journal **948**, 19 (2023). https://doi.org/10.3847/1538-4357/acb7dd ; https://arxiv.org/abs/2210.08358 .

[6] R. Bird, N. Tan, S. V. Luedtke, S. L. Harrell, M. Taufer and B. Albright, *VPIC 2.0: Next Generation Particle-in-Cell Simulations*, IEEE Transactions on Parallel and Distributed Systems **33**, 952-963 (2022). https://doi.org/10.1109/TPDS.2021.3084795 ; https://arxiv.org/abs/2102.13133 .

[7] Y. Yang, S. Adhikari and W. H. Matthaeus, *Electron Dissipation and Electromagnetic Work*, Journal of Geophysical Research: Space Physics **129**, e2024JA033105 (2024). https://doi.org/10.1029/2024JA033105 .

[8] O. Pezzi, H. Liang, J. L. Juno, P. A. Cassak, C. L. Vasconez, L. Sorriso-Valvo, D. Perrone, S. Servidio, V. Roytershteyn, J. M. TenBarge and W. H. Matthaeus, *Dissipation measures in weakly collisional plasmas*, Monthly Notices of the Royal Astronomical Society **505**, 4857-4873 (2021). https://doi.org/10.1093/mnras/stab1516 ; https://arxiv.org/abs/2101.00722 .

[9] T. Kalin, *Computational evidence for First-passage attribution and regional particle energization in multiscale reconnection*, version 1.0.0, Zenodo (2026), software and data. https://doi.org/10.5281/zenodo.23774456 . Computational snapshot e3119e6ee98f6d4bfc760a7c851d33f90b3a7d6b.